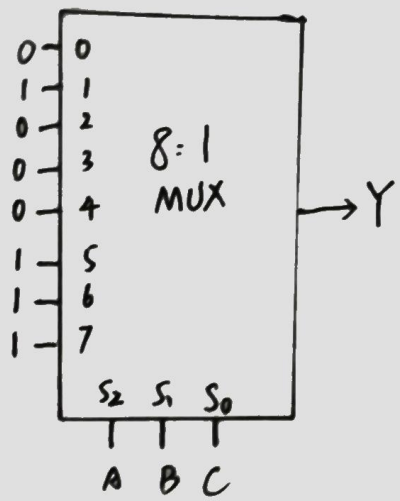
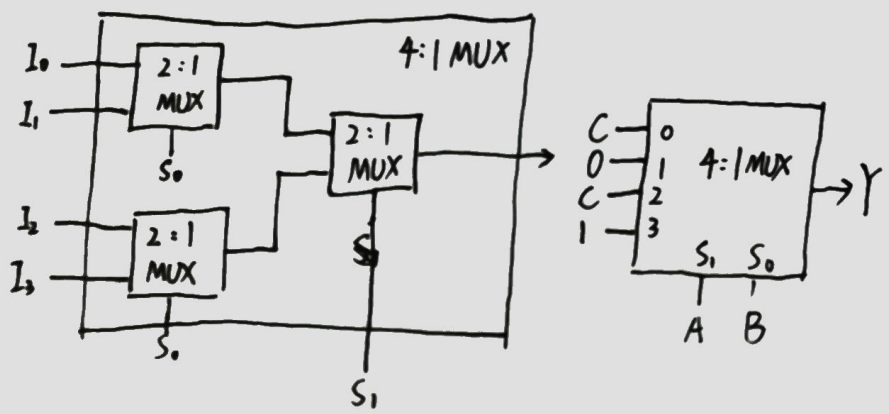


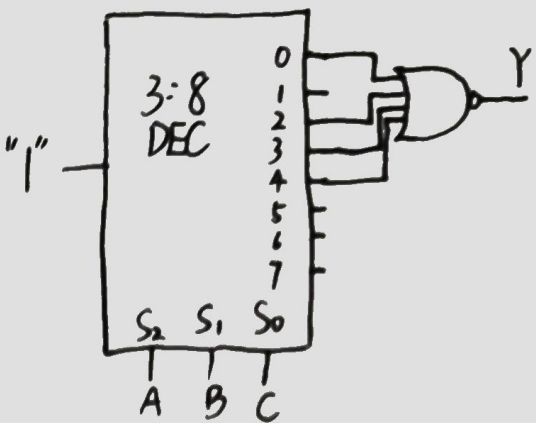
1. (a)



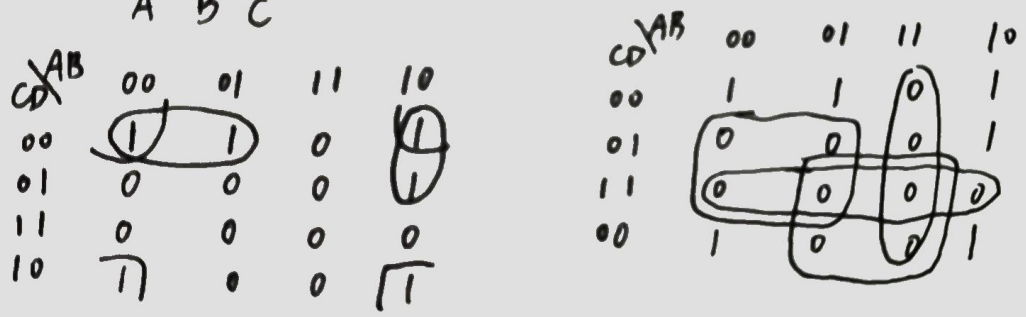
(b)



(c)

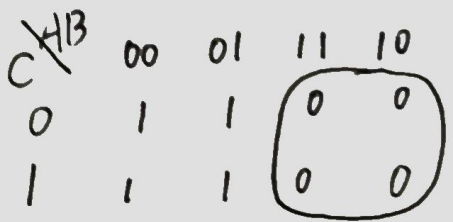


2.



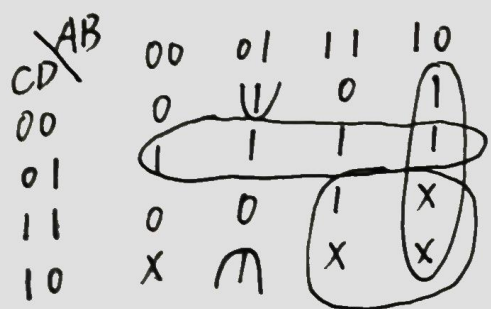
$$F(A, B, C, D) = B'D' + A'C'D' + AB'C' = (A' + B')(C' + D')(B' + C')(A + D')$$

3.



$$Y = A'$$

4.



$$Y = AC + AB' + C'D + A'BD'$$